

Programme	BSc Mathematics Honours			
Course Code	MAT8VN402			
Course Title	DATA VISUALIZATION			
Type of Course	Vocational Minor – Introduction to Data Science			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial	Practical	Total Hours
		per week	per week	
	4	3	2	75
Pre-requisites	Minor 1 and minor 2			
Course Summary	Course aims to provide data visualization techniques using R programming and interactive chart building			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the methods for visualizing data	U	C	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
CO2	Apply Visualization methods for different data domains	Ap	P	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
CO3	Design an Interactive data visualization story board for data	C	C	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Ext. Marks (70)
I		Introduction to Data Visualization	8	Min.10
	1	Definition, Methodology, Data Visualization and Theory, Visualization Design objectives		
	2	Key Factors – Purpose, visualization function and tone, visualization design options – Data representation, Data Presentation		
	3	Seven stages of data visualization, widgets, and introduction to different data visualization tools		
	4	Computational Statistics and Data Visualization, Presentation and Exploratory Graphics		
	5	Graphics and Computing, Statistical Historiography		
II		Visualizing Data Methods	13	Min.15
	6	Mapping, Time series, Connections and correlations - Scatter plot maps		
	7	Hierarchies and Recursion – introduction to Networks and Graphs, Info graphics		
	8	Complete Plots, Customization of plots -Parameters, Arranging Plots, Annotation,		
	9	Extensibility-Building Blocks, Combining Graphical Elements, 3-D Plots, Data Handling		
	10	Data and Graphs, Graph Layout Techniques, Graph Drawing		
	11	Bipartite Graphs, Hierarchical Trees, Spanning Trees, Networks, Directed Graphs, Tree maps		
III		Data visualization using R	12	Min.20
	12	Environment setup - R and RStudio, Basic plotting functions in R		
	13	Creating scatter plots, histograms, pie chat, bar charts, Boxplot, violin plot, line chart, heatmap, Customizing plot appearance,		
	14	Introduction to ggplot2, Grammar of graphics, creating static plots with ggplot2, Customizing plots with themes and scales		

	15	Introduction to plotly for interactive plotting, Creating interactive scatter plots, line plots, and bar charts, Adding interactivity with tooltips, zooming, and brushing		
	16	Designing interactive dashboards with Shiny and plotly, Other Visualization Packages		
IV		Introduction to Tableau	12	Min.15
	17	Environment Setup, Design flow, Data Types, File Types		
	18	Data Source - Custom Data View, Extracting Data, Field operations, Metadata, Data Joining and Blending		
	19	Worksheets- Adding, renaming, reordering Worksheet, Workbook Calculations		
	20	Sort and Filters- Sorting, Quick filtering, Context filtering, Condition filtering, Filter operations		
	21	Tableau Charts — Bar Chart, Line Chart, Multiple Measure Line Chart, Pie Chart		
	22	Scatter Plot, Bubble Chart, Bullet Graph, Box Plot, Dashboard – Formatting – Forecasting – Trend Lines		
		Practical's using R	30	
	1	Exploring Data with Basic Plots • Load a dataset (e.g., Iris dataset) into R. • Create scatter plots, histograms, and box plots to explore the distribution of variables. • Label axes, add titles, and customize colors and styles		
	2	Visualizing Relationships • Choose a dataset with multiple variables. • Create scatter plots to visualize relationships between pairs of variables. • Use color or shape to represent categorical variables. • Analyze patterns and correlations in the data		

	3	Time Series Visualization • Load a time series dataset (e.g., stock prices, weather data) into R. • Create line plots to visualize trends and fluctuations over time. • Use different line styles or colors to represent multiple time series. • Add labels, titles, and annotations to the plot		
	4	Bar and Pie Charts: • Load a dataset with categorical variables (e.g., survey responses, product categories). • Create bar charts and pie charts to visualize the distribution of categories. • Customize the appearance of the charts (e.g., colors, labels, legends).		
	5	Heatmaps and Correlation Plots: • Load a dataset with numerical variables (e.g., correlation matrix). • Create heatmaps to visualize correlations between variables. • Customize the color scheme and add annotations to the heatmap. • Interpret the patterns of correlation in the data		
	6	Box Plots and Violin Plots: • Load a dataset with numerical and categorical variables (e.g., Iris dataset). • Create box plots and violin plots to visualize the distribution of numerical variables across different categories. • Compare the use of box plots and violin plots for data visualization		

	7	Interactive Visualizations with ggplot2 and Shiny: • Create interactive plots using ggplot2 and Shiny. • Design a Shiny app with interactive controls (e.g., sliders, checkboxes) to explore different aspects of the data.		
	8	Geospatial Visualization: • Load a dataset with geographical information (e.g., map coordinates, regions). • Create maps using packages like ggmap, leaflet, or tmap to visualize spatial data. • Add layers, markers, and tooltips to the map to provide additional information		
	9	Faceted Plots: • Load a dataset with multiple groups or categories. • Create faceted plots using ggplot2 to display subsets of the data in separate panels. • Customize the appearance of each panel (e.g., axis limits, labels, titles		
	10	Network Visualization: • Load a dataset representing a network or graph (e.g., social network, co-authorship network). • Create network visualizations using packages like igraph or networkD3. • Customize the layout, node colors, and edge weights to convey information about the network structure.		
	11	Word Clouds and Text Visualization: • Load a dataset containing text data (e.g., tweets, reviews). • Create word clouds to visualize word frequency and importance. • Customize the appearance of the word cloud (e.g., colors, fonts, word sizes).		

	12	Dashboards with Plotly and Shiny: • Design an interactive dashboard using Plotly and Shiny. • Incorporate interactive plots, tables, and controls to explore and analyze data dynamically.		
	13	Dynamic Visualizations • Load a dataset with time-varying data (e.g., stock prices, sensor readings). • Create animated plots using package plotly. • Customize the animation settings (e.g., frame rate, transition effects) to enhance data visualization.		
	14	Visualizing Hierarchical Data • Load a dataset with hierarchical or nested structure (e.g., organizational hierarchy, file directories). • Create tree maps, dendrograms, or sunburst plots to visualize hierarchical data structures. • Customize the appearance of the plots to highlight different levels of hierarchy.		
	15	Dashboard Design • Design a dashboard layout with multiple visualizations and interactive components. • Arrange the visualizations in a coherent and informative manner. • Add text annotations, titles, and summaries to provide context and insights.		
		References		
	1	Ben Fry, “Visualizing Data”, O’Reilly Media, Inc., 2007.		
	2	Scott Murray, “Interactive data visualization for the web”, O’Reilly Media, Inc., 2nd edition, 2017		
	3	Fundamentals of Data Visualization" by Claus O. Wilke		
	4	Data Visualization: A Practical Introduction" by Kieran Healy		
	5	Learning tableau by Joshua N. Milligan		

Note: Proofs of all the results are exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	3	2	3	2	3	1	2
CO 2	3	2	3	2	3	2	3	1	2
CO 3	2	3	3	2	3	2	3	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓

MINOR COURSES